APEX_GLOBAL

COMPLETE

SYS_START: 06/04/2026 13:44:42 LAST_SYNC: 06/04/2026 16:41:52

Agentic Network Forensic Report – Apex Global Logistics

Executive Summary

This report was generated by a constrained forensic agent that uses alert-driven evidence extraction and a bounded reasoning loop. Its findings should be reviewed by a human analyst before external distribution.

Scope and Assumptions

Apex Global Logistics suffered a ransomware incident attributed to the Lynx group.

The available evidence is a 9-day PCAP captured from a core-switch tap. There is no

endpoint telemetry or memory image, so all conclusions must be grounded in network data.

Evidence Sources

- Alert JSON: `/Users/aaron/Desktop/Network_Security_Project/network/34936-sensor-alert-1a30b94e-4fe6-4f94-9d7e-c289ba1c7dcc.json`
- Zeek JSON: `/Users/aaron/Desktop/Network_Security_Project/network/34936-sensor-zeek-b0032190-fe13-4910-aae9-2907793ab566.json`

- PCAP dir: `/Users/aaron/Desktop/Network_Security_Project/network/pcap`

Ingest Statistics

METRIC	VALUE
Total Suricata alerts	161,663
Alerts – c2	10,000
Alerts – trojan	10,000
Alerts – other	10,000
Alerts – policy	6,559
Alerts – lateral	356
Alerts – scan	18
Zeek lines scanned	9,336,453
Zeek records matched	9,304,351
Total PCAPs indexed	117
Targeted PCAPs opened	30
DNS queries extracted	216
TLS sessions extracted	5000
HTTP requests extracted	0
SMB sessions extracted	0
RDP sessions extracted	0

Top Alert Rules

RULE	COUNT
ThreatFox botnet C2 traffic (domain - confidence level: 100%)	111930

ET MALWARE Known Hostile Domain ilo.brenz .pl Lookup	24329
ET DYN_DNS DNS Query to DynDNS Domain *.ufcfan .org	10585
ETPRO INFO Observed DNS Query to Dynamic DNS Service	6557
ET MALWARE UNK_FlappyBird Domain in DNS Lookup (onnetmais .org)	1057
ET DNS Reply Sinkhole - sinkhole.cert.pl 148.81.111.111	893
ET USER_AGENTS Microsoft Device Metadata Retrieval Client User-Agent	755
ET DNS Query to a .tk domain - Likely Hostile	587
ETPRO POLICY Observed DNS Query to Known ScreenConnect/ConnectWise Remote Deskto	576
ET REMOTE_ACCESS Observed DNS Query to Known ScreenConnect/ConnectWise Remote De	576

Targeted PCAP Files

- `34936-sensor-250305-00002430_redacted.pcap`
- `34936-sensor-250308-00002461_redacted.pcap`
- `34936-sensor-250308-00002473_redacted.pcap`
- `34936-sensor-250309-00002481_redacted.pcap`
- `34936-sensor-250309-00002488_redacted.pcap`
- `34936-sensor-250305-00002429_redacted.pcap`
- `34936-sensor-250303-00002384_redacted.pcap`
- `34936-sensor-250304-00002412_redacted.pcap`
- `34936-sensor-250304-00002393_redacted.pcap`
- `34936-sensor-250309-00002490_redacted.pcap`
- `34936-sensor-250303-00002386_redacted.pcap`
- `34936-sensor-250306-00002439_redacted.pcap`
- `34936-sensor-250309-00002491_redacted.pcap`
- `34936-sensor-250304-00002418_redacted.pcap`
- `34936-sensor-250304-00002402_redacted.pcap`
- `34936-sensor-250308-00002455_redacted.pcap`

- `34936-sensor-250308-00002464_redacted.pcap`
- `34936-sensor-250309-00002493_redacted.pcap`
- `34936-sensor-250307-00002447_redacted.pcap`
- `34936-sensor-250308-00002463_redacted.pcap`
- ... plus 10 more

Findings

A. Initial Access

Status: confirmed **Confidence:** HIGH **MITRE:** T1133

The host 10.128.239.57 is the most likely patient zero. Initial access was gained via Remote Desktop Protocol (RDP) on port 3389. Evidence shows 10.128.239.57 received a high volume of RDP connections (over 60,000 entries in zeek_rdp) from 164 distinct external IP addresses, suggesting a brute-force or credential stuffing attack. RDP cookies reveal attempted usernames such as 'Administr', 'Admin', 'WATER\Adm', and 'WATER\Use'. Following this initial access, 10.128.239.57 exhibited significant internal network activity, including 72,670 DCE/RPC operations (NetrLogonSamLogonEx) to internal hosts, notably 10.128.239.29 (a likely domain controller), indicating credential harvesting or lateral movement. Furthermore, 10.128.239.57 triggered 208 'ET INFO SMB2 NT Create AndX Request For an Executable File' alerts, attempting to create executable files on other internal hosts (e.g., 10.128.239.36, 10.128.239.34), which is a strong indicator of malware deployment and lateral movement.

Evidence:

- `ts?` | RDP | ` → 10.128.239.57 | High volume of RDP connections from external IPs to 10.128.239.57, indicating brute-force/credential stuffing. | _Artifact: zeek_rdp_`
- ts? | RDP | → 10.128.239.57 | RDP cookies showing attempted usernames. | _Artifact: zeek_rdp_
- ts? | DCERPC | 10.128.239.57 → 10.128.239.29 | Extensive internal DCE/RPC activity from 10.128.239.57 to other internal hosts, including a domain controller, indicating lateral movement/credential harvesting. | _Artifact: zeek_dce_rpc_

- ts? | SMB | 10.128.239.57 → 10.128.239.36 | Alerts indicating SMB2 executable file creation attempts from 10.128.239.57 to other internal hosts, suggesting malware deployment. | _Artifact: alerts_
- ts? | SMB | 10.128.239.57 → 10.128.239.34 | Alerts indicating SMB2 executable file creation attempts from 10.128.239.57 to other internal hosts, suggesting malware deployment. | _Artifact: alerts_

Limitations:

- The zeek_conn table is empty, preventing analysis of RDP session duration and byte transfers.
- The pcap_rdp table is empty, limiting deeper RDP session analysis.

Recommended next steps:

- Investigate the specific RDP credentials that were successfully used.
- Analyze the content of the executable files attempted to be created via SMB2.
- Trace the lateral movement path from 10.128.239.57 to other compromised hosts.
- Examine endpoint logs on 10.128.239.57 for process execution and user activity, if available.

B. Lateral Movement and Discovery

Status: agent_error Confidence: LOW MITRE: T1046, T1021.002

The agent was unable to complete its investigation: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in

9.208108081s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '9s'}]}}

Evidence:

- No evidence items recorded

Limitations:

- Agent error: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in 9.208108081s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '9s'}]}}

Recommended next steps:

- None recorded

C. Exfiltration

Status: agent_error Confidence: LOW MITRE: T1567

The agent was unable to complete its investigation: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in 8.826449676s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '8s'}]}}

Evidence:

- No evidence items recorded

Limitations:

- Agent error: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-

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limit: 5, model: gemini-2.5-flash\nPlease retry in 8.826449676s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '8s'}]}}

Recommended next steps:

- None recorded

D. Payload Deployment

Status: agent_error Confidence: LOW MITRE: T1021.001, T1021.002

The agent was unable to complete its investigation: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in 8.475576923s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model':

'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '8s'}]}}

Evidence:

- No evidence items recorded

Limitations:

- Agent error: 429 RESOURCE_EXHAUSTED. {'error': {'code': 429, 'message': 'You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemini-api/docs/rate-limits. To monitor your current usage, head to: https://ai.dev/rate-limit. \n* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 5, model: gemini-2.5-flash\nPlease retry in 8.475576923s.', 'status': 'RESOURCE_EXHAUSTED', 'details': [{'@type': 'type.googleapis.com/google.rpc.Help', 'links': [{'description': 'Learn more about Gemini API quotas', 'url': 'https://ai.google.dev/gemini-api/docs/rate-limits'}]}, {'@type': 'type.googleapis.com/google.rpc.QuotaFailure', 'violations': [{'quotaMetric': 'generativelanguage.googleapis.com/generate_content_free_tier_requests', 'quotaId': 'GenerateRequestsPerMinutePerProjectPerModel-FreeTier', 'quotaDimensions': {'location': 'global', 'model': 'gemini-2.5-flash'}, 'quotaValue': '5'}]}, {'@type': 'type.googleapis.com/google.rpc.RetryInfo', 'retryDelay': '8s'}]}}

Recommended next steps:

- None recorded

Timeline

- ts? | Initial Access | → 10.128.239.57 | High volume of RDP connections from external IPs to 10.128.239.57, indicating brute-

force/credential stuffing.

- ts? | Initial Access | → 10.128.239.57 | RDP cookies showing attempted usernames.
- ts? | Initial Access | 10.128.239.57 → 10.128.239.29 | Extensive internal DCE/RPC activity from 10.128.239.57 to other internal hosts, including a domain controller, indicating lateral movement/credential harvesting.
- ts? | Initial Access | 10.128.239.57 → 10.128.239.36 | Alerts indicating SMB2 executable file creation attempts from 10.128.239.57 to other internal hosts, suggesting malware deployment.
- ts? | Initial Access | 10.128.239.57 → 10.128.239.34 | Alerts indicating SMB2 executable file creation attempts from 10.128.239.57 to other internal hosts, suggesting malware deployment.

Methodology and Guardrails

The agent follows a four-phase alert-driven pipeline: (1) Suricata alert ingestion, (2) Zeek JSON correlation filtered to IOC IPs, (3) targeted PCAP selection by timestamp, (4) deep tshark analysis on selected PCAPs only.

- Read-only PCAP handling
- Fixed tool allowlist (initial_access, lateral_movement, exfiltration, payload_delivery)
- Evidence-linked conclusions – every finding cites actual log entries
- Confidence labels (HIGH/MEDIUM/LOW) with explicit limitations
- Safe abstention when evidence is weak

- `Results cached in` ingest/` to avoid re-scanning the 30 GB Zeek file

Cost and Efficiency Notes

- The Zeek JSON is streamed once and matched records are cached; re-runs are fast.
- tshark is invoked only on the targeted PCAP subset, not all 129 files.
- Deterministic mode has zero model cost; LLM mode adds cost only for planning.
- This keeps the expensive reasoning layer narrow and auditable.